MySQL — Zero to Master

22 topics, grouped as Beginner → Intermediate → Advanced → Master. Work through the topics in order; type every code example yourself and finish the practice tasks.

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Beginner Level

01. Introduction to Databases and MySQL

DEFINITION

A database stores data in an organised way. MySQL is a popular open-source relational database that stores data in tables (rows and columns) and is queried with SQL.

SYNTAX

```
mysql -u root -p
```

EXAMPLE

```
mysql -u root -p
SHOW DATABASES;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A big ledger room where every shop's accounts are kept in tidy tables; MySQL stores data in rows and columns and answers questions with SQL.

தமிழ்

ஒவ்வொரு கடையின் கணக்குகளும் ஒழுங்கான அட்டவணைகளில் இருக்கும் பெரிய பதிவறை போல MySQL தரவை வரிசை, நெடுவரிசைகளில் சேமித்து SQL கேள்விகளுக்குப் பதில் தருகிறது.

PROS

- Free and widely used
- Fast for read-heavy apps
- Easy to learn
- Great tool support

CONS

- Fewer advanced SQL features than PostgreSQL
- Historically weaker standards compliance
- Oracle-controlled

ADVANCED / NEW FEATURES

- MySQL 8.4 LTS
- Innovation releases 9.x with VECTOR data type
- Atomic DDL, CTEs, window functions (8.0)
- `utf8mb4` default

02. Installation and Basic Commands

DEFINITION

Install MySQL Server and a client (CLI, Workbench, DBeaver). Create and select a database.

SYNTAX

```
CREATE DATABASE name;
USE name;
```

EXAMPLE

```
CREATE DATABASE school;
USE school;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Setting up a new shop's cupboard before keeping goods; you create a database and select it before making tables.

தமிழ்

பொருட்களை வைக்கும் முன் புதிய அலமாரி அமைப்பது போல tables உருவாக்கும் முன் database உருவாக்கி தேர்ந்தெடுக்கிறோம்.

PROS

- Simple setup
- Many GUI tools (Workbench, DBeaver)

CONS

- Default configs not production tuned

ADVANCED / NEW FEATURES

- MySQL Shell and Router
- Docker images for quick setup

03. Data Types and Creating Tables

DEFINITION

`INT`, `BIGINT`, `DECIMAL`, `VARCHAR`, `TEXT`, `DATE`, `DATETIME`, `BOOLEAN`, `JSON`. Define keys and constraints.

SYNTAX

```
CREATE TABLE t (id INT PRIMARY KEY, name VARCHAR(100) NOT NULL);
```

EXAMPLE

```
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE,
  dob DATE
);
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A ration card register with fixed columns: card number, head of family, address. Each column has a defined data type.

தமிழ்

அட்டை எண், குடும்பத் தலைவர், முகவரி என நிலையான நெடுவரிசைகள் உள்ள ரேஷன் அட்டைப் பதிவேடு போல ஒவ்வொரு நெடுவரிசைக்கும் தரவு வகை உண்டு.

PROS

- Strong structure
- Right types save space

CONS

- Schema changes need care on big tables

ADVANCED / NEW FEATURES

- Invisible indexes
- `JSON` type
- Instant `ADD COLUMN` (8.0)

04. INSERT, SELECT, UPDATE, DELETE (CRUD)

DEFINITION

The four basic operations. Always use `WHERE` with `UPDATE` and `DELETE`.

SYNTAX

```
INSERT INTO t (a) VALUES (1);
SELECT * FROM t WHERE ...;
UPDATE t SET a=1 WHERE ...;
DELETE FROM t WHERE ...;
```

EXAMPLE

```
INSERT INTO students (name, email) VALUES ('Arun', 'arun@x.com');
SELECT * FROM students WHERE id = 1;
UPDATE students SET name = 'Arun K' WHERE id = 1;
DELETE FROM students WHERE id = 1;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A shop ledger: add a sale, look it up, correct it, or cancel it. Always say which entry (WHERE) before changing.

தமிழ்

கடைப் பதிவேடு: விற்பனையைச் சேர், தேடு, திருத்து, ரத்து செய்; மாற்றும் முன் எந்தப் பதிவு (WHERE) என்று சொல்ல வேண்டும்.

PROS

- Simple and standard
- Fast for typical operations

CONS

- UPDATE/DELETE without WHERE changes everything
- Large deletes are slow

ADVANCED / NEW FEATURES

- `INSERT ... ON DUPLICATE KEY UPDATE`
- Safe-updates mode

05. Filtering and Sorting

DEFINITION

`WHERE`, `AND/OR/NOT`, `IN`, `BETWEEN`, `LIKE`, `IS NULL`, `ORDER BY`, `LIMIT`, `DISTINCT`.

SYNTAX

```
SELECT cols FROM t WHERE cond ORDER BY col LIMIT n;
```

EXAMPLE

```
SELECT name FROM students WHERE name LIKE 'A%' ORDER BY name LIMIT 10;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Asking the record clerk 'show only Chennai customers sorted by name, first ten'; filtering and sorting narrow results.

தமிழ்

'சென்னை வாடிக்கையாளர்களை மட்டும், பெயர் வரிசையில், முதல் பத்து' என்று எழுத்தரிடம் கேட்பது போல filtering, sorting முடிவுகளைக் குறைக்கும்.

PROS

- Expressive filtering
- `LIKE`, `IN`, `BETWEEN`

CONS

- `LIKE '%x'` cannot use indexes
- OFFSET paging is slow on big tables

ADVANCED / NEW FEATURES

- Better optimiser hints
- Descending indexes (8.0)

06. Constraints and Keys

DEFINITION

`PRIMARY KEY`, `FOREIGN KEY`, `UNIQUE`, `NOT NULL`, `DEFAULT`, `CHECK`.

SYNTAX

```
PRIMARY KEY (id)
FOREIGN KEY (c) REFERENCES t(id)
```

EXAMPLE

```
ALTER TABLE marks ADD FOREIGN KEY (student_id) REFERENCES students(id);
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Aadhaar number is unique to one person (primary key); a school marks sheet refers to a student ID (foreign key).

தமிழ்

ஆதார் எண் ஒருவருக்கே உரியது (primary key); பள்ளி மதிப்பெண் தாள் மாணவர் ID-ஐக் குறிப்பிடும் (foreign key).

PROS

- Data integrity
- Prevents orphan records

CONS

- Constraints slow bulk loads slightly

ADVANCED / NEW FEATURES

- `CHECK` constraints enforced (8.0.16)

Intermediate Level

07. Aggregate Functions and GROUP BY

DEFINITION

`COUNT`, `SUM`, `AVG`, `MIN`, `MAX` with `GROUP BY` and `HAVING`.

SYNTAX

```
SELECT g, COUNT(*) FROM t GROUP BY g HAVING COUNT(*) > 1;
```

EXAMPLE

```
SELECT city, COUNT(*) AS total FROM students GROUP BY city HAVING total > 5;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A shopkeeper totals daily sales by product; aggregates give counts, sums and averages per group.

தமிழ்

கடைக்காரர் தினசரி விற்பனையை பொருள் வாரியாகக் கூட்டுவது போல aggregates குழுவுக்கு count, sum, average தரும்.

PROS

- Powerful reporting
- Standard functions

CONS

- Wrong GROUP BY gives confusing results

ADVANCED / NEW FEATURES

- `ONLY_FULL_GROUP_BY` default
- Window functions as alternative

08. Joins

DEFINITION

`INNER`, `LEFT`, `RIGHT` joins combine related tables; understand cross and self joins.

SYNTAX

```
SELECT ... FROM a JOIN b ON a.id = b.a_id;
```

EXAMPLE

```
SELECT s.name, m.score FROM students s
JOIN marks m ON m.student_id = s.id;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Combining the customer register with the orders register using customer ID to see who ordered what.

தமிழ்

யார் எதை ஆர்டர் செய்தார் என்று பார்க்க வாடிக்கையாளர் பதிவேடு, ஆர்டர் பதிவேட்டை customer ID மூலம் இணைப்பது போல.

PROS

- Combines normalised data
- Index-friendly

CONS

- Missing indexes make joins slow
- Many joins are hard to read

ADVANCED / NEW FEATURES

- Hash join (8.0.18)
- Hash join for all joins (8.0.20)

09. Subqueries and CTEs

DEFINITION

Queries inside queries; Common Table Expressions (`WITH`) improve readability.

SYNTAX

```
WITH x AS (SELECT ...) SELECT * FROM x;
```

EXAMPLE

```
WITH top AS (SELECT student_id, AVG(score) a FROM marks GROUP BY student_id)
SELECT * FROM top WHERE a > 80;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A rough work-sheet used to compute something before writing the final answer; subqueries and CTEs break big questions into steps.

தமிழ்

இறுதி விடை எழுதும் முன் கணக்கிட பயன்படும் கரடு தாள் போல subqueries, CTEs பெரிய கேள்வியை படிகளாகப் பிரிக்கின்றன.

PROS

- Readable queries
- Recursive CTEs for trees

CONS

- Can be slower if not optimised

ADVANCED / NEW FEATURES

- CTEs and recursive CTEs (8.0)

10. Normalization and Schema Design

DEFINITION

1NF, 2NF, 3NF remove duplication; model one-to-many and many-to-many with foreign keys and junction tables.

EXAMPLE

```
CREATE TABLE enrollments (student_id INT, course_id INT, PRIMARY KEY (student_id, course_id));
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Keeping customer details in one register and orders in another instead of copying addresses on every bill; normalisation avoids duplicate data.

தமிழ்

ஒவ்வொரு பில்லிலும் முகவரியை நகலெடுக்காமல் வாடிக்கையாளர் விவரத்தை ஒரு பதிவேட்டிலும் ஆர்டர்களை மற்றொன்றிலும் வைப்பது போல normalisation நகல் தரவைத் தவிர்க்கும்.

PROS

- Less duplication
- Easier updates

CONS

- More joins
- Over-normalising hurts performance

ADVANCED / NEW FEATURES

- Generated columns and JSON help hybrid designs

11. Built-in Functions

DEFINITION

String, date, numeric, conditional (`CASE`, `IF`, `COALESCE`) functions.

SYNTAX

```
SELECT NOW(), CONCAT(a, b), COALESCE(x, 0);
```

EXAMPLE

```
SELECT CONCAT(first, ' ', last), DATEDIFF(NOW(), dob) FROM students;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A calculator and date stamp at the counter; built-in functions format and compute values for you.

தமிழ்

கவுண்டரில் உள்ள கால்குலேட்டர், தேதி முத்திரை போல built-in functions மதிப்புகளை வடிவமைத்து கணக்கிடுகின்றன.

PROS

- Rich function library
- Faster than app-side loops

CONS

- Functions on indexed columns prevent index use

ADVANCED / NEW FEATURES

- JSON functions
- Regex functions (8.0)

12. Views and Altering Tables

DEFINITION

Views save queries; `ALTER TABLE` changes structure.

SYNTAX

```
CREATE VIEW v AS SELECT ...;
ALTER TABLE t ADD COLUMN c INT;
```

EXAMPLE

```
CREATE VIEW top_students AS SELECT * FROM students WHERE gpa > 9;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A saved report format that anyone can open; views save complex queries and ALTER changes structure.

தமிழ்

யார் வேண்டுமானாலும் திறக்கக்கூடிய சேமிக்கப்பட்ட அறிக்கை வடிவம் போல views சிக்கலான queries-ஐச் சேமிக்கின்றன.

PROS

- Reusable queries
- Hide complexity

CONS

- Views can hide slow queries

ADVANCED / NEW FEATURES

- Instant DDL for many ALTERs

Advanced Level

13. Indexes and Query Optimisation

DEFINITION

Indexes speed up reads; use `EXPLAIN` to inspect plans; avoid functions on indexed columns and `SELECT *`.

SYNTAX

```
CREATE INDEX idx ON t (col);
EXPLAIN SELECT ...;
```

EXAMPLE

```
CREATE INDEX idx_email ON students(email);
EXPLAIN SELECT * FROM students WHERE email = 'a@x.com';
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A book's index at the back lets you find a topic instantly instead of reading every page; database indexes work the same way.

தமிழ்

புத்தகத்தின் பின் பக்க அட்டவணை ஒவ்வொரு பக்கமும் படிக்காமல் தலைப்பைக் கண்டுபிடிக்க உதவுவது போல database indexes செயல்படுகின்றன.

PROS

- Huge read speed-ups
- `EXPLAIN` shows plans

CONS

- Slow writes and extra storage
- Too many indexes hurt

ADVANCED / NEW FEATURES

- Invisible and descending indexes
- `EXPLAIN ANALYZE` (8.0.18)

14. Transactions and ACID

DEFINITION

Group statements so all succeed or none do. InnoDB supports ACID and row-level locking; know isolation levels.

SYNTAX

```
START TRANSACTION; ... COMMIT;
```

EXAMPLE

```
START TRANSACTION;
UPDATE acc SET bal = bal - 100 WHERE id = 1;
UPDATE acc SET bal = bal + 100 WHERE id = 2;
COMMIT;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Transferring money: deduct from one account and add to another; either both happen or none (ACID).

தமிழ்

பணம் மாற்றும்போது ஒரு கணக்கில் குறைத்து மற்றொன்றில் சேர்ப்பது; இரண்டும் நடக்க வேண்டும் அல்லது எதுவும் நடக்கக்கூடாது (ACID).

PROS

- Data consistency
- Row-level locking in InnoDB

CONS

- Long transactions block others
- Deadlocks possible

ADVANCED / NEW FEATURES

- Atomic DDL (8.0)
- `NOWAIT` and `SKIP LOCKED`

15. Stored Procedures, Functions, Triggers

DEFINITION

Server-side logic: procedures, functions, triggers and events.

SYNTAX

```
CREATE PROCEDURE p() BEGIN ... END;
```

EXAMPLE

```
DELIMITER //
CREATE PROCEDURE get_student(IN sid INT) BEGIN SELECT * FROM students WHERE id = sid; END //
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A standard recipe the cook follows every time; stored routines keep logic inside the database.

தமிழ்

சமையல்காரர் ஒவ்வொரு முறையும் பின்பற்றும் நிலையான செய்முறை போல stored routines logic-ஐ database-இலேயே வைக்கின்றன.

PROS

- Reusable server-side logic
- Less network traffic

CONS

- Harder to version and test
- Vendor specific syntax

ADVANCED / NEW FEATURES

- Triggers and events improvements
- JavaScript stored programs in Enterprise

16. Window Functions

DEFINITION

`ROW_NUMBER`, `RANK`, `SUM() OVER`, `LAG` for analytics without collapsing rows.

SYNTAX

```
SELECT RANK() OVER (ORDER BY score DESC) FROM t;
```

EXAMPLE

```
SELECT name, score, RANK() OVER (ORDER BY score DESC) FROM marks;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A school rank list: every student sees their rank while the full list stays visible; window functions rank without collapsing rows.

தமிழ்

பள்ளி தரவரிசைப் பட்டியலில் எல்லா மாணவர்களின் வரிசை தெரிவது போல window functions வரிசைகளைக் குறைக்காமல் rank தருகின்றன.

PROS

- Powerful analytics
- Cleaner than self-joins

CONS

- Can be heavy on large sets

ADVANCED / NEW FEATURES

- Window functions added in MySQL 8.0

17. JSON and Full-text Search

DEFINITION

JSON columns with `->>`; full-text indexes for search.

SYNTAX

```
SELECT data->>'$.city' FROM t;
```

EXAMPLE

```
SELECT data->>'$.city' FROM profiles;
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A form with free-text remarks beside fixed columns; JSON columns store flexible data inside a SQL table.

தமிழ்

நிலையான நெடுவரிசைகளுடன் கூடுதல் குறிப்புகள் உள்ள படிவம் போல JSON நெடுவரிசைகள் SQL table-க்குள் நெகிழ்வான தரவைச் சேமிக்கின்றன.

PROS

- Flexible schema parts
- Multi-valued indexes

CONS

- Weaker than a document database for heavy JSON use

ADVANCED / NEW FEATURES

- Multi-valued indexes
- JSON_TABLE (8.0.4)

18. Users, Privileges and Security

DEFINITION

Create least-privilege users, avoid SQL injection with prepared statements, enable TLS, rotate passwords.

SYNTAX

```
CREATE USER 'a'@'h' IDENTIFIED BY '...';
GRANT SELECT ON db.* TO 'a'@'h';
```

EXAMPLE

```
CREATE USER 'app'@'%' IDENTIFIED BY '...';
GRANT SELECT, INSERT ON shop.* TO 'app'@'%';
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Giving shop staff keys only to the rooms they need; least-privilege users protect the database.

தமிழ்

பணியாளர்களுக்கு தேவையான அறைகளுக்கு மட்டும் சாவி கொடுப்பது போல குறைந்த அனுமதி பயனர்கள் database-ஐப் பாதுகாக்கின்றனர்.

PROS

- Fine-grained permissions
- Roles (8.0)

CONS

- Over-privileged accounts are common

ADVANCED / NEW FEATURES

- Roles
- `caching_sha2_password` default
- Password policies

Master Level

19. Backup and Restore

DEFINITION

Logical backups with `mysqldump`; binary logs for point-in-time recovery.

SYNTAX

```
mysqldump db > backup.sql
```

EXAMPLE

```
mysqldump -u root -p shop > shop.sql
mysql -u root -p shop < shop.sql
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A duplicate key and a copy of the ledger kept in a bank locker; backups let you recover after disaster.

தமிழ்

வங்கி லாக்கரில் வைக்கும் நகல் சாவி, பதிவேட்டு நகல் போல backups பேரழிவுக்குப் பின் மீட்க உதவுகின்றன.

PROS

- Simple logical backups
- Binary logs for point-in-time recovery

CONS

- mysqldump is slow for huge data

ADVANCED / NEW FEATURES

- MySQL Shell dump/load utilities
- Clone plugin
- XtraBackup

20. Replication and High Availability

DEFINITION

Source-replica replication, read scaling, failover, Group Replication / InnoDB Cluster.

SYNTAX

```
CHANGE REPLICATION SOURCE TO ...
```

EXAMPLE

```
SHOW REPLICA STATUS\G
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A branch shop keeps a copy of the head office stock list so customers can be served even if the head office is busy; replicas copy the primary.

தமிழ்

தலைமை அலுவலகம் பிஸியாக இருந்தாலும் வாடிக்கையாளருக்கு சேவை செய்ய கிளைக் கடை இருப்புப் பட்டியலின் நகலை வைத்திருப்பது போல replicas primary-ஐ நகலெடுக்கின்றன.

PROS

- Read scaling
- Failover support

CONS

- Replication lag
- Operational complexity

ADVANCED / NEW FEATURES

- InnoDB Cluster and Group Replication
- `SOURCE` / `REPLICA` terminology (8.0.22+)

21. Partitioning and Scaling

DEFINITION

Partition large tables, shard horizontally, use connection pooling and caching.

SYNTAX

```
PARTITION BY RANGE (YEAR(d))
```

EXAMPLE

```
PARTITION BY RANGE (YEAR(created)) (...)
```

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

Old bills kept year-wise in separate files so searching is quick; partitioning splits huge tables.

தமிழ்

பழைய பில்களை ஆண்டுவாரியாக தனி கோப்புகளில் வைத்து தேடலை விரைவாக்குவது போல partitioning பெரிய tables-ஐப் பிரிக்கிறது.

PROS

- Faster queries on large data
- Easy archiving

CONS

- Partition key limitations
- Not a substitute for indexes

ADVANCED / NEW FEATURES

- Native partitioning in InnoDB
- HeatWave for analytics

22. Real Project: Library / E-commerce Schema

DEFINITION

Design tables, relations, indexes and reports (top sellers, monthly revenue) with sample data and stored procedures.

REAL-WORLD EXAMPLE (ENGLISH + தமிழ்)

English

A complete library or online store schema with sample data and reports is the standard practice project.

தமிழ்

மாதிரித் தரவு, அறிக்கைகளுடன் கூடிய முழுமையான நூலகம் அல்லது ஆன்லைன் கடை schema ஒரு நிலையான பயிற்சி project.

PROS

- Real-world design practice

CONS

- Time consuming

ADVANCED / NEW FEATURES

- Use MySQL 8.4 LTS with Docker for practice

Interview Question Bank

Q1. DELETE vs TRUNCATE vs DROP?
DELETE removes rows (can filter, rollbackable); TRUNCATE empties the table fast; DROP removes the table itself.

Q2. INNER JOIN vs LEFT JOIN?
INNER returns matching rows; LEFT returns all left rows plus matches (NULL when none).

Q3. WHERE vs HAVING?
WHERE filters rows before grouping; HAVING filters groups after aggregation.

Q4. What is normalization?
Organising tables to reduce redundancy and anomalies (1NF-3NF).

Q5. What is an index?
A data structure (B-tree) that speeds up lookups at the cost of extra writes/storage.

Q6. What is ACID?
Atomicity, Consistency, Isolation, Durability - guarantees for transactions.

Q7. PRIMARY KEY vs UNIQUE?
Primary key identifies rows, is not null and only one per table; UNIQUE can be many and allows NULLs.

Q8. What is a foreign key?
A column referencing another table's key to enforce referential integrity.

Q9. How do you find slow queries?
Slow query log, `EXPLAIN`, Performance Schema.

Q10. What is SQL injection?
Attack that inserts SQL through input; prevented by prepared statements.

Capstone Project Ideas

1. Student management schema
2. Library database
3. E-commerce orders and reports
4. Hospital appointment system
5. Banking transactions